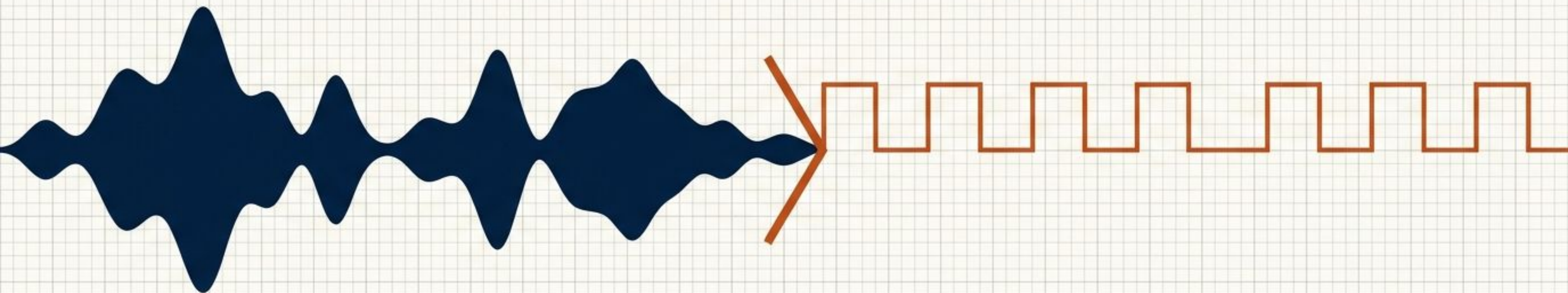


BOUNDING THE UNPREDICTABLE

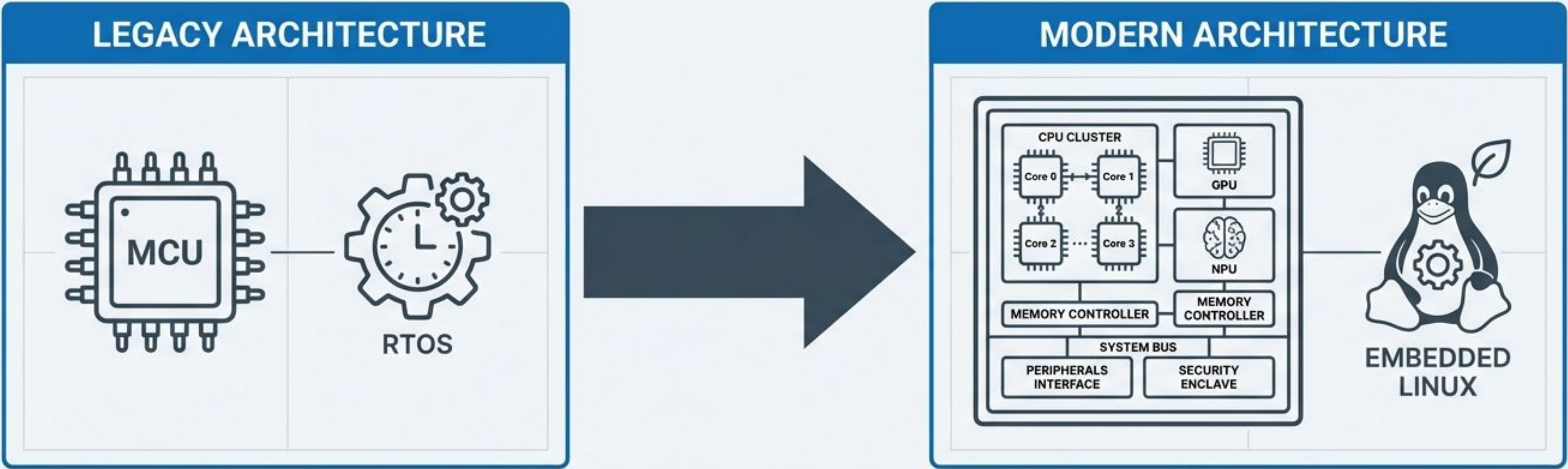
Deterministic Performance of Real-Time Audio on Linux SoCs

Yasin Arslan & Suleyman Sertac Ust (Hacettepe University)



Characterizing and bounding deterministic audio latency
in safety-critical communication systems.

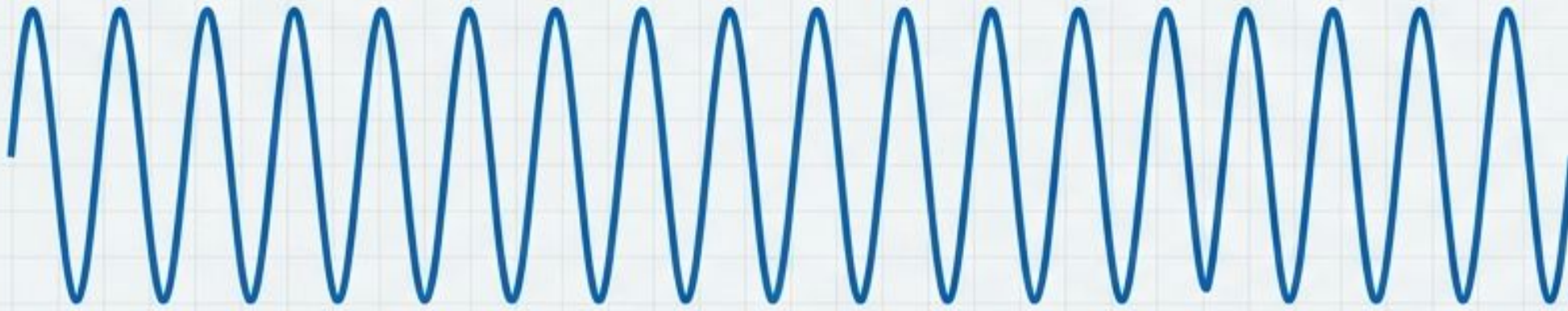
1. INTRODUCTION: THE TECHNOLOGY SHIFT



DRIVERS OF THE TRANSITION & MISSION-CRITICAL APPLICATIONS		
Performance: Massively higher compute power	Ecosystem: Rich, mature software availability & architectural flexibility	Target Use Cases: Defense systems, emergency communications, industrial control

2. THE PROBLEM: REAL-TIME AUDIO ON LINUX

THE REQUIREMENT



- **Ultra-low** absolute latency
- Low jitter (variance)
- **Strict, deterministic** timing guarantees

THE LINUX REALITY

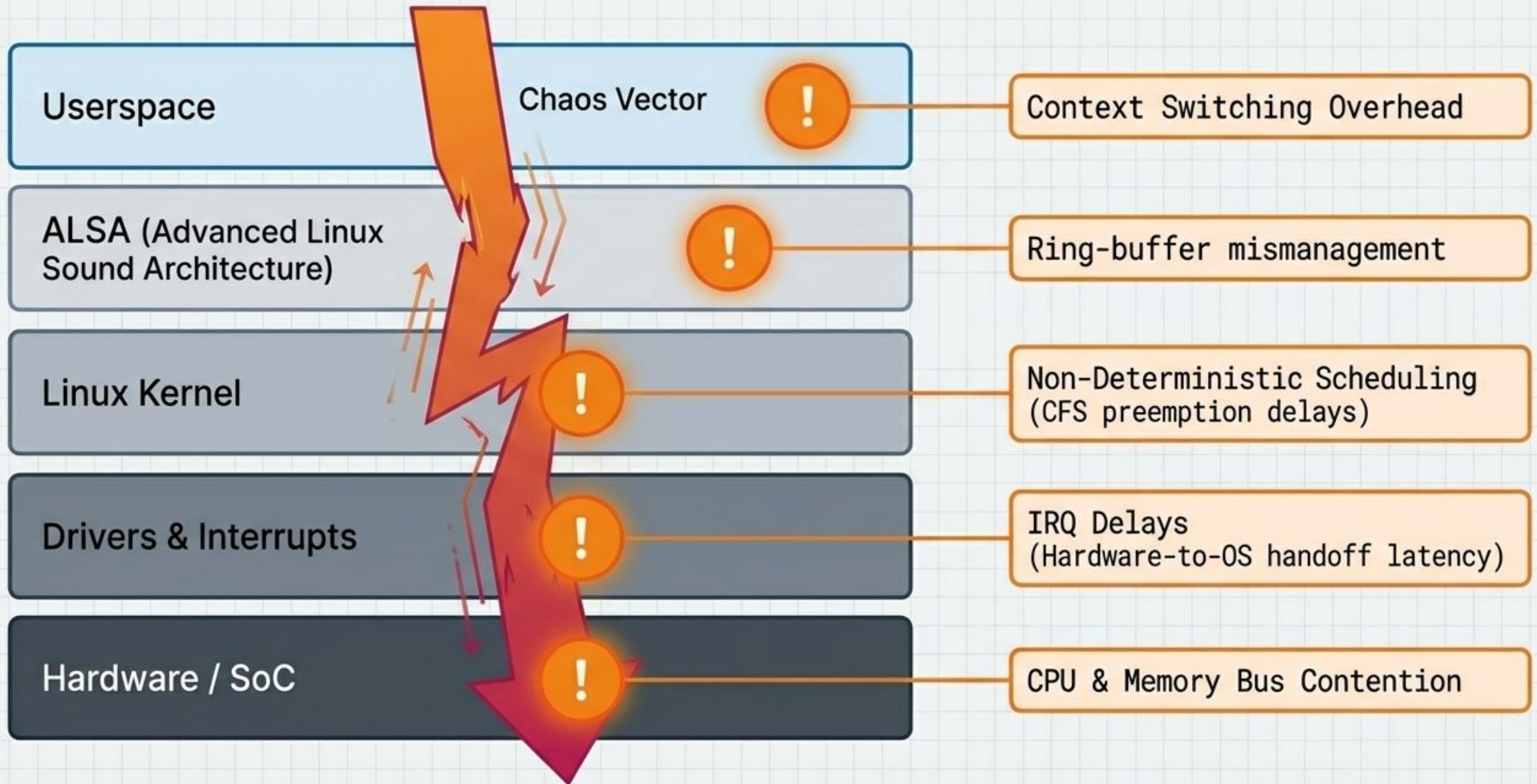


Completely Fair Scheduler (CFS) Flaws

- Standard Linux is inherently **NON-DETERMINISTIC**.
- Optimizes for overall throughput, ignoring strict execution timing.
- Results in **variable latency**, severe jitter, and missed audio deadlines.

Analogy: A competitive online game with a 30ms average ping, but sudden 200ms lag spikes.

3. ROOT CAUSES: THE ACCUMULATION OF DELAY



4. EXISTING SOLUTIONS & THEIR LIMITATIONS

THE PREEMPT_RT PATCH

- Improves latency via enforced kernel preemptibility.
- **Limitation:** Reduces, but cannot guarantee, absolute determinism under heavy system load.

CURRENT RESEARCH METHODOLOGIES

- Focuses heavily on kernel-only metrics (e.g., cyclicttest).
- **Limitation:** Analyzes average latency, ignoring worst-case tail latency.

THE CRITICAL GAP

- No end-to-end application pipeline analysis.
- Lack of **predictive latency modeling**.
- Primarily hardware-based (costly, opaque, and difficult to reproduce).

5. THE RESEARCH GAP

There is currently NO systematic, simulation-based framework that:



Measures strict
END-TO-END audio
application latency.



Models latency
sources
ANALYTICALLY.



Validates predictions
under **CONTROLLED,**
REPRODUCIBLE
stress conditions.

6. PROPOSED APPROACH: THE DUAL METHODOLOGY

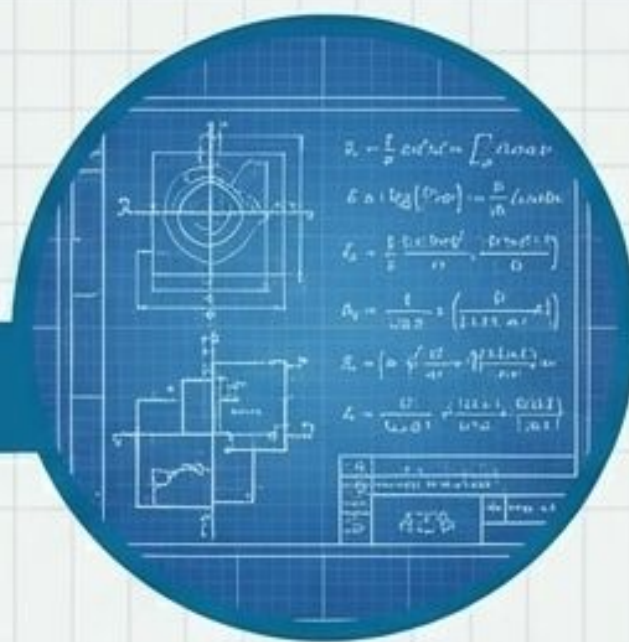


EXPERIMENTAL MEASUREMENT

Rigorous empirical benchmarking of the true end-to-end audio flow.



VALIDATED DETERMINISM

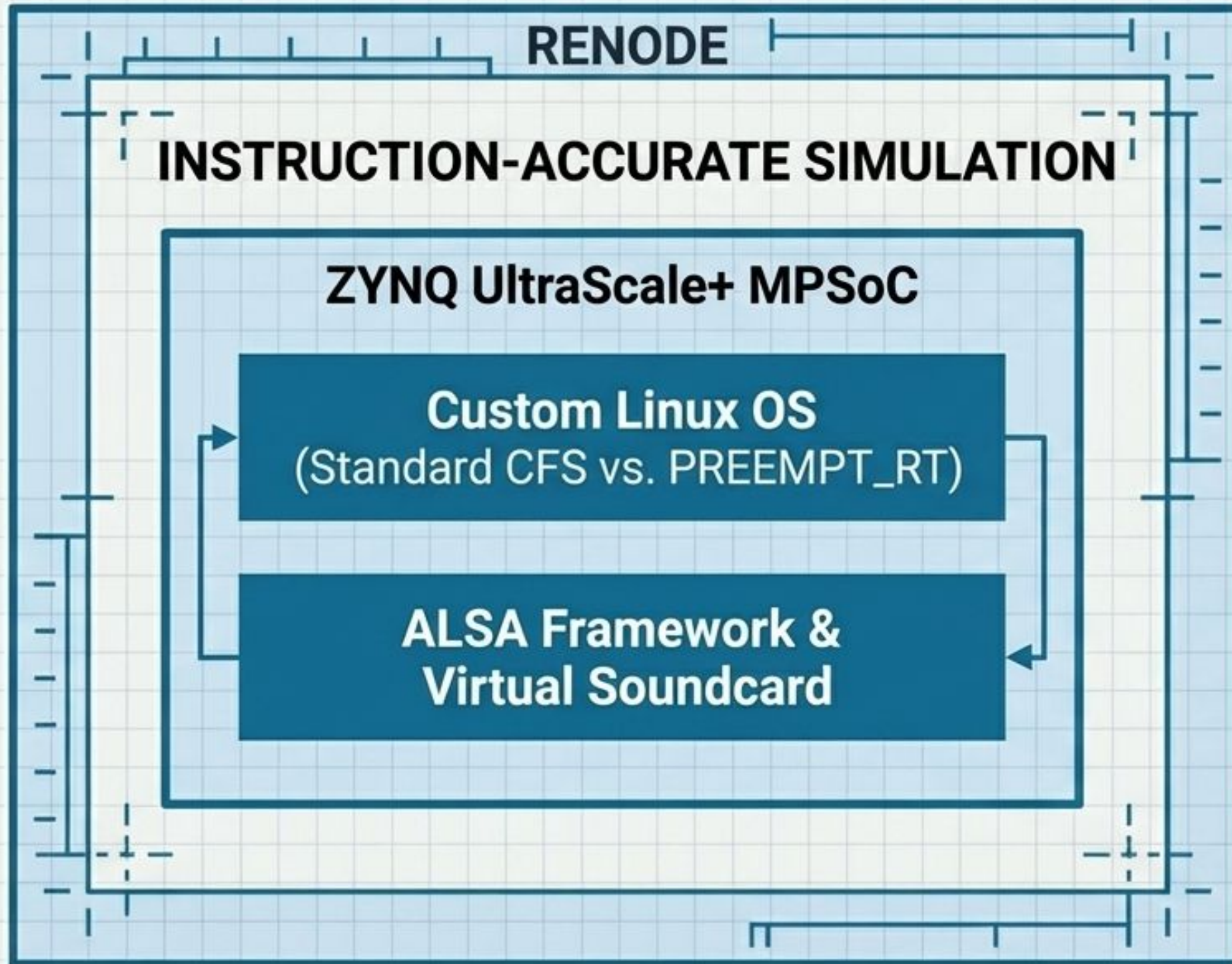


PREDICTIVE LATENCY MODEL

Mathematical budgeting of all architectural delay sources.

THE CORE PHILOSOPHY: We don't just **measure** latency to see what happens. We analytically **EXPLAIN** every **microsecond of delay**.

7. SYSTEM OVERVIEW: SIMULATED DETERMINISM



WHY SIMULATION?



100% Reproducibility
(No hardware noise)

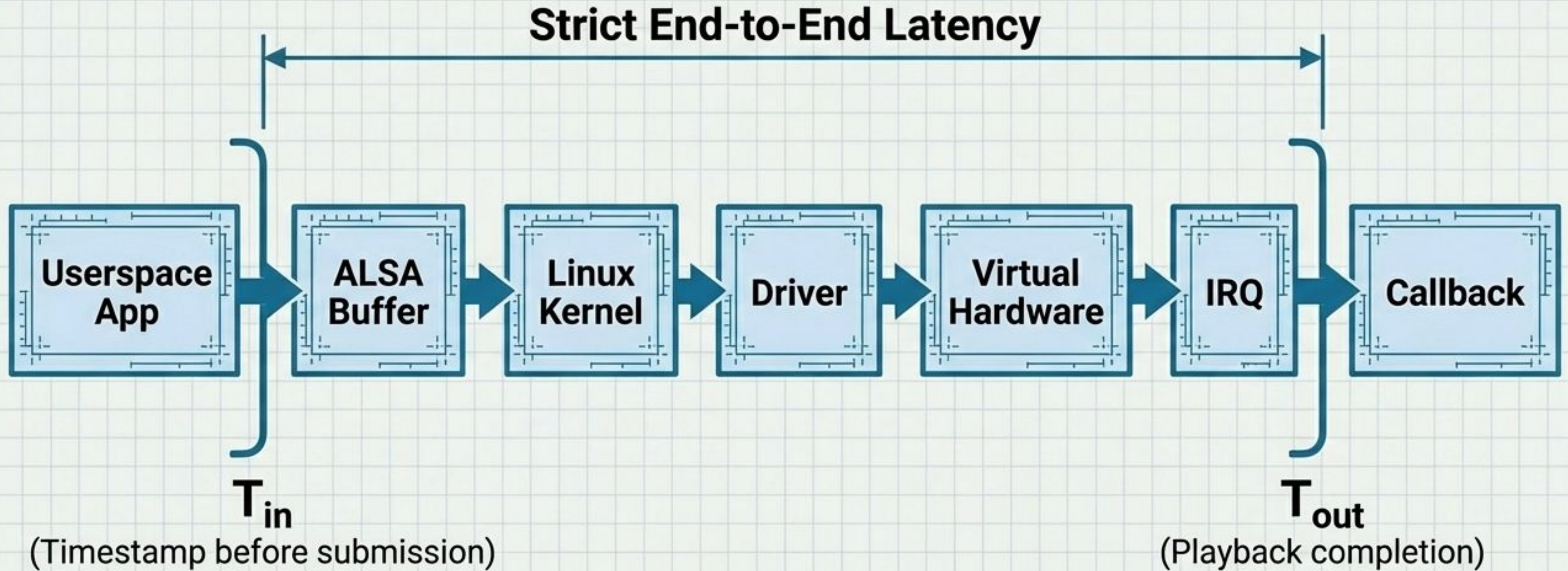


Total Controllability
of Virtual Time

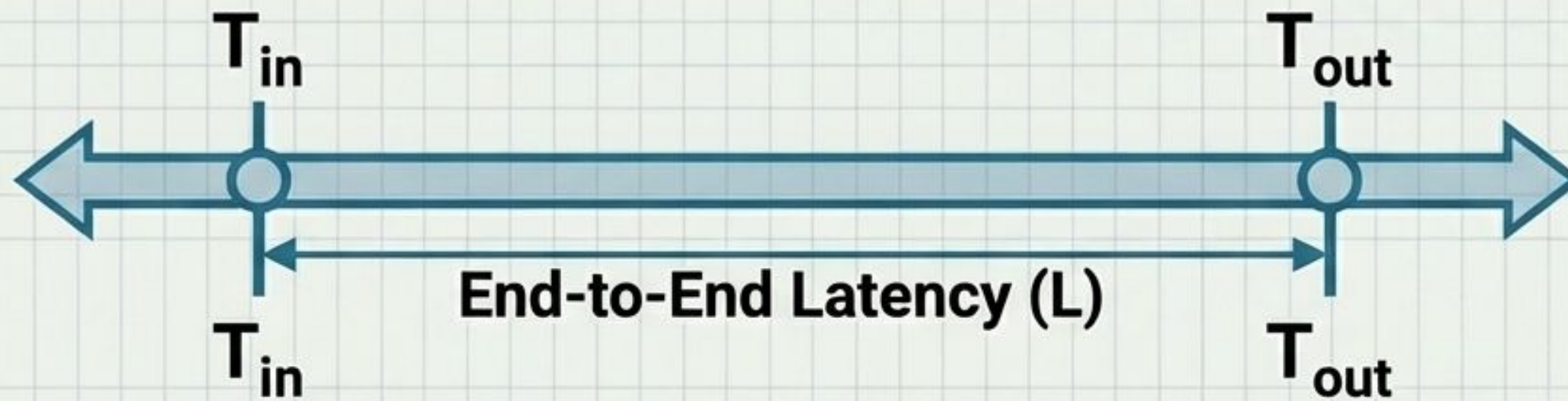


Hardware-Independent
Cost Efficiency

8. AUDIO PIPELINE: TRACING THE DATA FLOW



9. LATENCY & JITTER DEFINITIONS



$$L = T_{out} - T_{in}$$

$$J_{max} = \max(L) - \min(L)$$

Measured over a 100-frame sliding window

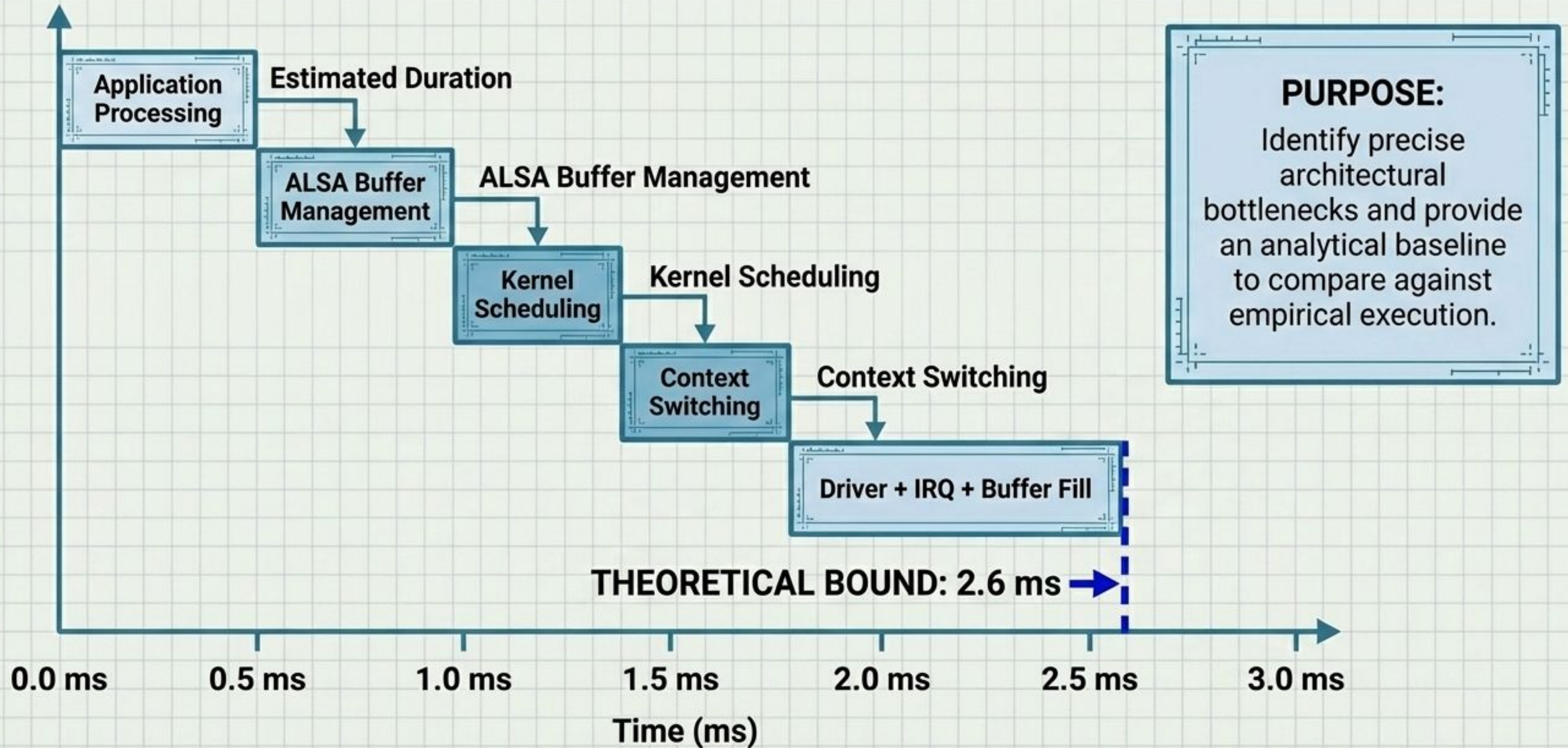
STRICT DETERMINISM THRESHOLDS

$L_{max} = 10 \text{ ms}$
(Hard deadline for interactive voice)

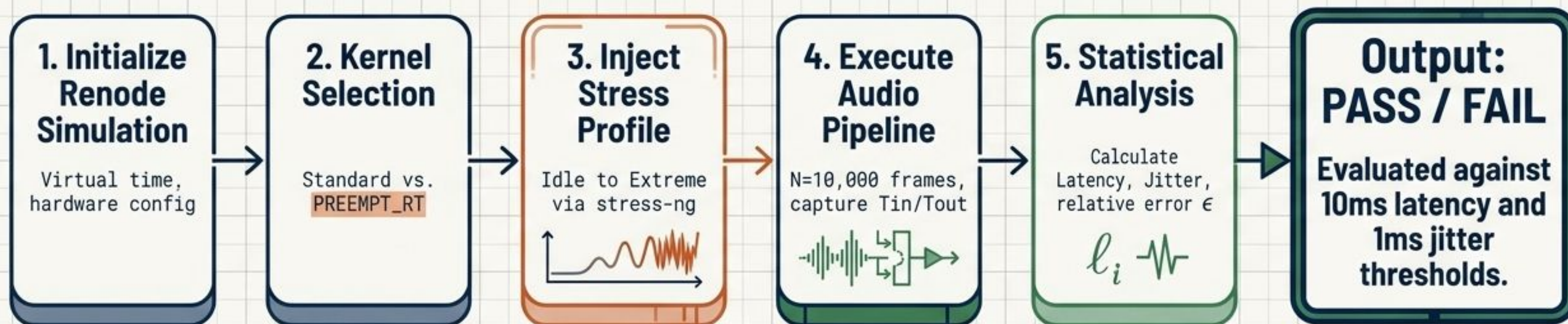
$J_{th} = 1 \text{ ms}$
(Maximum allowable variance)

A single frame exceeding L_{max} equals a deadline miss.

10. THE PREDICTIVE LATENCY BUDGET

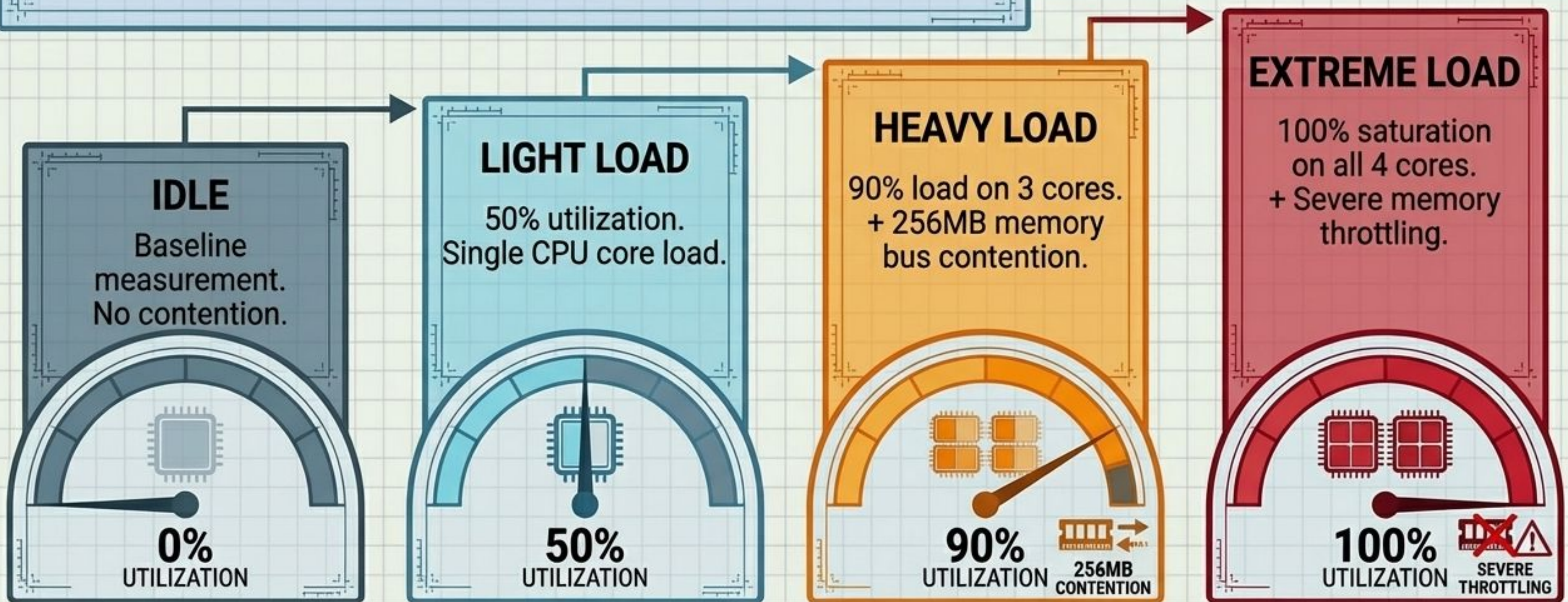


RENODE AUDIO PIPELINE EVALUATION FLOW



12. SYNTHETIC STRESS SCENARIOS

WHY? To simulate real-world, chaotic operational environments like interrupt storms and heavy parallel computing.



13. EVALUATION METRICS



LATENCY PROFILE

Tracking Mean, Maximum, and the 99th Percentile delay. Focuses on tail latency, as averages hide failures.



WORST-CASE JITTER

Peak latency variance. Continuously monitored over sliding time windows.



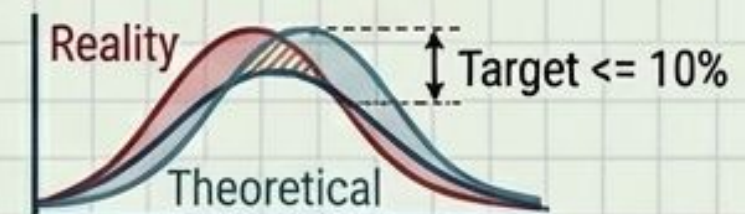
DEADLINE MISS RATE (DMR)

The exact percentage of frames breaching the 10 ms strict safety threshold.

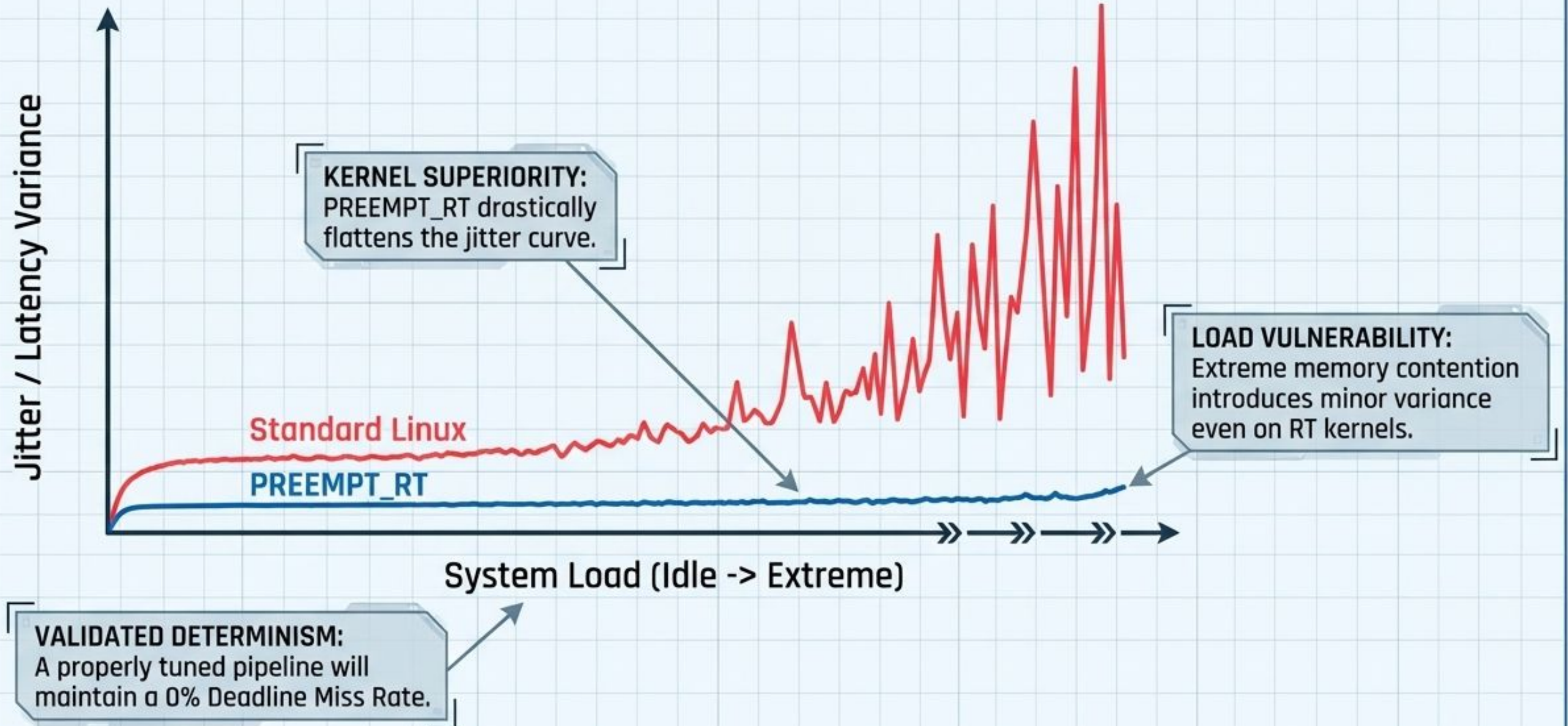


MODEL ERROR (Epsilon)

Variance between experimental reality and the predictive theoretical budget. Target: Must be $\leq 10\%$.



14. EXPECTED RESULTS



15. CORE CONTRIBUTIONS

SIMULATION FRAMEWORK

Hardware-independent audio testing natively integrated into the Renode environment.



END-TO-END ANALYSIS

A permanent shift away from isolated kernel metrics to absolute application-level truth.



PREDICTIVE MODEL

A mathematically validated latency budget for complex Linux audio pipelines.

$$p^t = \frac{1}{2\pi} (xy - z)^2$$



REPRODUCIBLE EXPERIMENTS

Fully scriptable environments ensuring transparent, repeatable science expensive physical prototypes.



16. SYSTEM LIMITATIONS

SIMULATION VS. PHYSICAL REALITY

Emulation provides deterministic virtual time, but cannot capture complex thermal throttling or physical electrical delays.



DRAM CONTENTION ABSTRACTION

Renode does **not currently** model cycle-accurate memory bus physics, requiring reliance on synthetic software stress.



PERIPHERAL SIMPLIFICATION

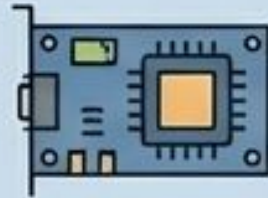
The virtual **soundcard behaves perfectly** according to spec, which may mask idiosyncratic, buggy behaviors found in real-world hardware drivers.



17. FUTURE WORK: TRAJECTORIES

REAL HARDWARE VALIDATION

Deploying the exact pipeline to physical Zynq MPSoC silicon.



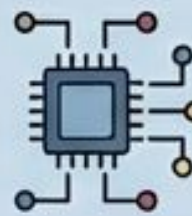
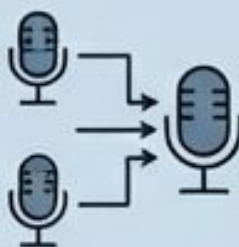
ENERGY OPTIMIZATION

Correlating real-time CPU scheduling with power consumption and Dynamic Voltage Scaling (DVFS).



MULTI-CHANNEL SCALABILITY

Expanding architecture from mono audio to complex multi-channel microphone arrays.



SECURITY IMPROVEMENTS



Implementing kernel-space buffer isolation to prevent side-channel attacks.

18. CONCLUSION

THE REALITY:

- The embedded industry's shift to Linux is inevitable and powerful, but standard Linux is inherently non-deterministic.



THE REQUIREMENT:

- Mission-critical audio demands strict, predictable timing guarantees; a late packet is a lost packet.



THE SOLUTION:

- By combining an **analytical mathematical latency budget** with a **strictly controlled simulation environment**, this methodology provides a *reproducible framework* to guarantee real-time performance on heterogeneous SoCs.

